

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Direct & Inverse Proportion

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

DIRECT & INVERSE PROPORTION · CH1 · L15

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The Map

This lesson collapses Direct & Inverse Proportion into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Choose direct or inverse

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Find the constant k

TRIGGER *"work out", "find", a missing value*

03 Substitute the new value

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Check the direction of change

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Find inverse function

TRIGGER *"work out", "find", a missing value*

06 Use when

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 16 website questions, 16 checked solutions, 3 local PDFs read.

01 Choose direct or inverse

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Direct & Inverse Proportion question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$A = \frac{k}{C^2}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1H Q15

Find inverse function

$$A = \frac{k}{C^2}$$

Use when

$$40 = \frac{k}{1.5^2}$$

Finish: $C = 0.3$.

FORMULA BANK

Direct: $y = kx$. Inverse: $y = \frac{k}{x}$. Find k first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

15 A is inversely proportional to C
A = 40 when C = 1.5
Calculate the value of C when A = 1000
C =

YOUR SOLUTION

02 Find the constant k

TRIGGER *"work out", "find", a missing value*

BECOMES A Direct & Inverse Proportion question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y \propto \frac{1}{\sqrt{x}}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P2HR Q20

Find inverse function

$$y \propto \frac{1}{\sqrt{x}}$$

Find inverse function

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: $T = \frac{100}{9}$.

FORMULA BANK

Direct: $y = kx$. Inverse: $y = \frac{k}{x}$. Find k first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 02.

QUESTION

20 y is inversely proportional to x x is directly proportional to T 3 Given that y = 8 when T = 25 find the exact value of T when y = 27 T =

YOUR SOLUTION

03 Substitute the new value

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Direct & Inverse Proportion question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$A = \frac{k}{C^2}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1H Q15

Find inverse function

$$A = \frac{k}{C^2}$$

Use when

$$40 = \frac{k}{1.5^2}$$

Finish: $C = 0.3$.

FORMULA BANK

Direct: $y = kx$. Inverse: $y = \frac{k}{x}$. Find k first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

04 Check the direction of change

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Direct & Inverse Proportion question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$M = kh^3$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P2H Q17

Find inverse function

$$M = kh^3$$

Use when

$$4 = k(0.5)^3$$

Finish: $h = 2.5$.

FORMULA BANK

Direct: $y = kx$. Inverse: $y = \frac{k}{x}$. Find k first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

17 M varies directly as the cube of h M = 4 when h = 0.5 Find the value of h when M = 500

YOUR SOLUTION

05 Find inverse function

TRIGGER *"work out", "find", a missing value*

BECOMES A Direct & Inverse Proportion question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$A = \frac{k}{C^2}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1H Q15

Find inverse function

$$A = \frac{k}{C^2}$$

Use when

$$40 = \frac{k}{1.5^2}$$

Finish: $C = 0.3$.

FORMULA BANK

Direct: $y = kx$. Inverse: $y = \frac{k}{x}$. Find k first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

06 Use when

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Direct & Inverse Proportion question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$A = \frac{k}{C^2}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
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WORKED MODEL

Source pattern: Jan 2022 P1H Q15

Find inverse function

$$A = \frac{k}{C^2}$$

Use when

$$40 = \frac{k}{1.5^2}$$

Finish: $C = 0.3$.

FORMULA BANK

Direct: $y = kx$. Inverse: $y = \frac{k}{x}$. Find k first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Choose direct or inverse
"work out", "find", a missing value	Find the constant k
"work out", "show", "hence", a familiar exam phrase	Substitute the new value
"work out", "show", "hence", a familiar exam phrase	Check the direction of change
"work out", "find", a missing value	Find inverse function
"work out", "show", "hence", a familiar exam phrase	Use when

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.